

Lecture 08: Diagnostics II

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- 1 Identifying outlying X observations: Hat matrix leverage values
- 2 Identifying influential cases: DFFITS, Cook's Distance, and DFBETAS Measures
 - Influence on single fitted value - DFFITS
 - Influence on all fitted values - Cook's distance
 - Influence on the Regression Coefficients - *DFBETAS*
- 3 Multicollinearity diagnostics: Variance inflation factor
 - Informal diagnostics
 - Formal diagnostics: Variance inflation diagnostics
 - Shrinkage methods
- 4 Surgical unit example

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Leverage and influential points

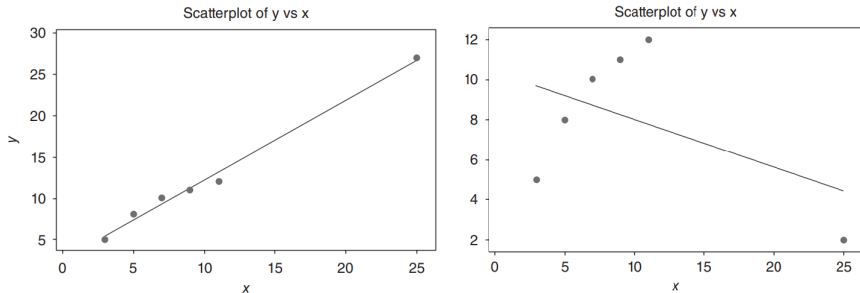


Figure: Outlier in X: Pure leverage (left) and influential (right) points.

Example of Outlying cases

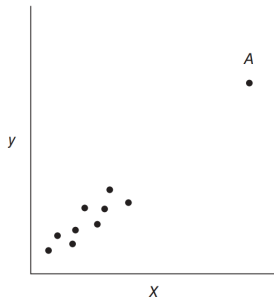


Figure 6.1 An example of a leverage point.

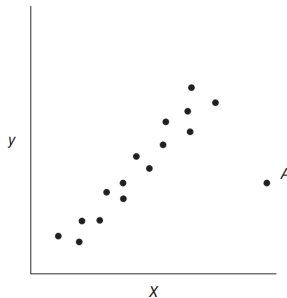
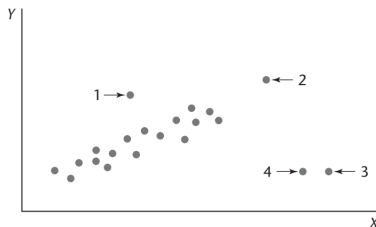


Figure 6.2 An example of an influential observation.

Example of Outlying cases



- Outlying causes dramatic effects on the fitted least squares regression function.
- Case 1 is outlying with respect to its Y value, given X ; cases 2, 3, and 4 are outlying with respect to their X values. Cases 3 and 4 are also outlying with respect to both Y and X . 3 and 4 may be influence points.
- When the model has multiple variables, identify outlying cases is not obvious. Some multivariable outliers may not be detectable in single-variable or two-variable analyses.

Use of hat matrix for identifying outlying X observations

The hat matrix also is helpful in directly identifying outlying X observations. Some properties of diagonal of $\mathbf{H}$ are as follows:

- $0 \leq h_{ii} \leq 1$, $\sum_{i=1}^n h_{ii} = p$, where p is the number of regression parameters in the regression function including the intercept term.
- h_{ii} is a measure of the distance between the X values for the i th case and the means of the X values for all n cases. Hence, h_{ii} is also called the **leverage** of the i th case.

Use of hat matrix for identifying outlying X observations

- If the i th case is outlying in terms of its X observations it has a large leverage value h_{ii} .
- A leverage value h_{ii} is usually considered to be large (evidence of an outlying case) if:
 - Since mean of leverage values is p/n , leverage values greater than $2p/n$ are considered outlying cases with regard to their X values.
 - h_{ii} values exceeding 0.5 indicate very high leverage, whereas those between 0.2 and 0.5 indicate moderate leverage.
 - The existence of a gap between the leverage values for most of the cases and the unusually large leverage value(s).

Body fat example revisit

Let's examine the Body Fat notebook.

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Why identifying influential cases is important?

- Recall: The cases whose studentized deleted residuals are large in absolute value is identifies as outlying Y observations.
- After identifying cases that are outlying with respect to their Y and/or their X , the next step is to ascertain whether or not these outlying cases are influential.
- A case to be influential if its exclusion causes major changes in the fitted regression function. Hence, it should be removed.
- Three measures of influence that are widely used in practice, each based on the omission of a single case to measure its influence:
 - 1 Influence on single fitted value - DFFITS
 - 2 Influence on all fitted values - Cook's Distance
 - 3 Influence on the regression coefficients - DFBETAS

Influence on single fitted value - DFFITS

$$(DFFITS)_i = \frac{\hat{Y}_i - \hat{Y}_{i(i)}}{\sqrt{MSE_{(i)} h_{ii}}},$$

where

- $\hat{Y}_i$ is fitted value for the i th case when all n cases are used in fitting the regression function
- $\hat{Y}_{i(i)}$ is the predicted value for the i th case obtained when the i th case is omitted in fitting the regression function
- $\sqrt{MSE_{(i)} h_{ii}}$ is the estimated standard deviation of $\hat{Y}_i$

It is proved that

$$(DFFITS)_i = e_i \left[\frac{n - p - 1}{SSE(1 - h_{ii}) - e_i^2} \right]^{1/2} \left(\frac{h_{ii}}{1 - h_{ii}} \right)^{1/2} = t_i \left(\frac{h_{ii}}{1 - h_{ii}} \right)^{1/2},$$

where t_i is the studentized deleted residual.

Recommend guideline:

A case is influential if the absolute value of DFFITS exceeds 1 for small to medium data sets and $2\sqrt{p/n}$ for large data sets.

Show the students the calculation of DFFITS, Body Fat dataset with two predictors.

Cook's distance

Cook's distance D_i is an aggregate influence measure, showing the effect of the i th case on all n fitted values:

$$D_i = \frac{\sum_{j=1}^n (\hat{Y}_j - \hat{Y}_{j(i)})^2}{p * MSE}$$

In matrix form

$$\mathbf{D} = \frac{(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)})'(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)})}{p * MSE},$$

where $\hat{\mathbf{Y}}$ is the vector of the fitted values when all n cases are used for the regression fit and $\hat{\mathbf{Y}}_{(i)}$ is the vector of the fitted values when the i th case is deleted.

- D_i is related to the $F(p, n - p)$ distribution and ascertain the corresponding percentile value
- If the percentile value is less than about 10 or 20 percent, the i th case has little apparent influence on the fitted values

Cook's distance can be calculated from

$$D_i = \frac{e_i^2}{pMSE} \left[\frac{h_{ii}}{(1 - h_{ii})^2} \right]$$

the i th case can be influential if:

- 1 having a large residual e_i and only a moderate leverage value h_{ii} , or
- 2 having a large leverage value h_{ii} with only a moderately sized residual e_i , or
- 3 having both a large residual e_i and a large leverage value h_{ii} .

Show the students the calculation of Cook's distance, Body Fat dataset with two predictors.

Influence on the Regression Coefficients - *DFBETAS*

- A measure of the influence of the i th case on each regression coefficient b_k ($k = 0, 1, \dots, p - 1$) is the difference between the estimated regression coefficient b_k based on all n cases and the regression coefficient obtained when the i th case is omitted, $b_{k(i)}$.
- *DFBETAS* is defined by the difference divided by an estimate of the standard deviation of b_k

$$(DFBETAS)_{k(i)} = \frac{b_k - b_{k(i)}}{\sqrt{MSE_i c_{kk}}}, k = 0, 1, \dots, p - 1,$$

where c_{kk} is the k th diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$.

Influence on the Regression Coefficients - *DFBETAS*

- A large absolute value of $(DFBETAS)_{k(i)}$ is indicative of a large impact of the i th case on the k th regression coefficient.
- It is recommended that A case influential if the absolute value of *DFBETAS* exceeds 1 for small to medium data sets and $2/\sqrt{n}$ for large data sets.

Let's revisit Body Fat with two predictors example.

Using the class https://www.statsmodels.org/stable/generated/statsmodels.stats.outliers_influence.OLSInfluence.html.

Usage example is at https://www.statsmodels.org/stable/examples/notebooks/generated/regression_diagnostics.html

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Multicollinearity

Typical problems when the predictor variables are highly correlated (multicollinearity):

- Adding or deleting a predictor variable changes the regression coefficients.
- The extra sum of squares associated with a predictor variable varies, depending upon which other predictor variables are already included in the model.
- The estimated standard deviations of the regression coefficients become large when the predictor variables in the regression model are highly correlated with each other.
- The estimated regression coefficients individually may not be statistically significant even though a definite statistical relation exists between the response variable and the set of predictor variables.

Informal diagnostics for Multicollinearity

- 1 Large changes in the estimated regression coefficients when a predictor variable is added or deleted, or when an observation is altered or deleted.
- 2 Nonsignificant results in individual tests on the regression coefficients for important predictor variables.
- 3 Estimated regression coefficients with an algebraic sign that is the opposite of that expected from theoretical considerations or prior experience.
- 4 Large coefficients of simple correlation between pairs of predictor variables in the correlation matrix r_{XX} .
- 5 Wide confidence intervals for the regression coefficients representing important predictor variables.

Informal diagnostics - Body fat example

Subject	Triceps Skinfold Thickness	Thigh Circumference	Midarm Circumference	Body Fat
i	X_{i1}	X_{i2}	X_{i3}	Y_i
1	19.5	43.1	29.1	11.9
2	24.7	49.8	28.2	22.8
3	30.7	51.9	37.0	18.7
...	...	...	...	...
18	30.2	58.6	24.6	25.4
19	22.7	48.2	27.1	14.8
20	25.2	51.0	27.5	21.1

The predictor variables, triceps skinfold thickness and thigh circumference, are highly correlated because of:

- The large changes in the estimated regression coefficients and their estimated standard deviations when a variable was added,
- Nonsignificant results in individual tests on anticipated important variables,
- An estimated negative coefficient when a positive coefficient was expected

Variance inflation factor

- The variance-covariance matrix of the estimated regression coefficients is given by:

$$\text{var}\{b\} = \sigma^2(X'X)^{-1}$$

- The variance-covariance matrix of the estimated *standardized* regression coefficients is:

$$\text{var}\{b^*\} = (\sigma^*)^2 r_{X'X}^{-1},$$

where $r_{X'X}$ is the matrix of the pairwise simple correlation coefficients among the X variables and $(\sigma^*)^2$ is the error term variance for the transformed model.

Variance inflation factor

- The diagonal element $(VIF)_k$ of the matrix $r_{X'X}^{-1}$ is called the *variance inflation factor (VIF)* for b_k^* .
- The variance of b_k^* ($k = 1, \dots, p - 1$) is equal to

$$\text{var}\{b_k^*\} = (\sigma^*)^2 (VIF)_k$$

- It is proved that

$$(VIF)_k = \frac{1}{1 - R_k^2}, k = 1, 2, \dots, p - 1, \text{ hence,}$$
$$\text{var}\{b_k^*\} = \frac{(\sigma^*)^2}{1 - R_k^2}$$

where R_k^2 is the coefficient of multiple determination when X_k is regressed on the other variables in the model.

Variance inflation factor and Multicollinearity

- When $R_k^2 = 0$ (X_k is not linearly related to the other X variables), then $(VIF)_k = 1$
- When $R_k^2 \neq 0$ (there is intercorrelations among the X variables), then $(VIF)_k > 1$.
- When $R_k^2 = 1$ (there is a perfect linear association with the other X variables in the model), then $(VIF)_k$ and $\sigma^2\{b_k^*\}$ are unbounded.

Variance inflation factor uses

- The largest *VIF* value among all *X* variables is often used as an indicator of the severity of multicollinearity.
- A maximum *VIF* value in excess of 10 is frequently taken as an indication that multicollinearity may be unduly influencing the estimates.
- It has been proved that

$$E\left\{\sum_{k=1}^{p-1}(b_k^* - \beta_k^*)^2\right\} = (\sigma^*)^2 \sum_{k=1}^{p-1}(VIF)_k$$

Hence, the mean of the *VIF* values $\frac{\sum_{k=1}^{p-1}(VIF)_k}{p-1}$ provides information about the severity of the multicollinearity in terms of how far the estimated standardized regression coefficients b_k^* are from the true values β_k . Mean *VIF* values considerably larger than 1 are indicative of serious multicollinearity problems.

Body fat example

Variable	b_k^*	$(VIF)_k$
X_1	4.2637	708.84
X_2	-2.9287	564.34
X_3	-1.5614	104.61
Maximum $(VIF)_k = 708.84$		$(\overline{VIF}) = 459.26$

- The expected sum of the squared errors in the least squares standardized regression coefficients is nearly 460 times as large as it would be if the X variables were uncorrelated.
- All three VIF values greatly exceed 10, which again indicates that serious multicollinearity problems exist.
- Note that $(VIF)_3 = 105$ despite the fact that both r_{13}^2 and r_{23}^2 are not large, which is an instance where X_3 is strongly related to X_1 and X_2 together ($R_{32} = 0.990$)

Shrinkage methods: Ridge regression

- 1 Ridge regression makes the assumption that the regression coefficients (after normalization) should not be very large.
- 2 We will look at some methods that allow us to shrink this additional information into a more useful form.
- 3 This is a reasonable assumption in applications where you have a large number of predictors and you believe that many of them have some effect on the response.
- 4 Ridge regression is particularly effective when the model matrix is collinear and the usual least squares estimates of β appear to be unstable.

Shrinkage methods: Ridge regression

- Minimize the penalized least square with L_2 -norm:

$$\sum_{i=1}^n (Y_i - (\beta_0 + \dots + \beta_{p-1}X_{i,p-1}))^2 + \lambda \sum_j \beta_j^2$$

Shrinkage methods: LASSO and Elastic Net

- LASSO: minimize the penalized least square with L_1 -norm:

$$\sum_{i=1}^n (Y_i - (\beta_0 + \dots + \beta_{p-1} X_{i,p-1}))^2 + \lambda \sum_j |\beta_j|$$

- ElasticNet: combining Ridge regression and LASSO
- See
https://www.statsmodels.org/dev/generated/statsmodels.regression.linear_model.OLS.fit_regularized.html

Outline

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Surgical unit example

Using Surgical unit notebook.

Summary

- Check model adequacy using added-variable plots
- Identify outliers in X , Y
- Identify influential cases: DFFITS, Cook's distance, and DFBETAS

- Chapter 10, Kutner's book
- https://www.statsmodels.org/stable/examples/notebooks/generated/regression_diagnostics.html
- https://www.statsmodels.org/stable/examples/notebooks/generated/regression_plots.html

THANK YOU FOR YOUR ATTENTION!